

1 DIAGNOSE



WorldEcho

Expert + Off-Expert Query

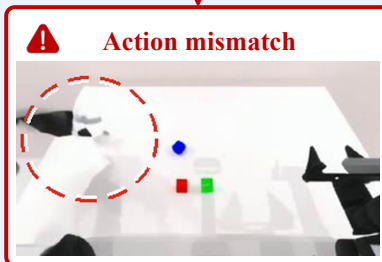
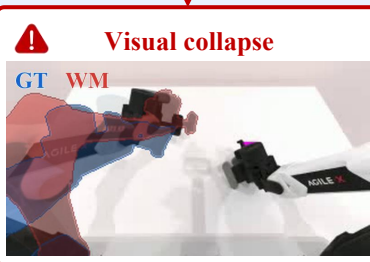
- Expert
- Cross-State Replay
- Policy Rollout
- Local Perturbation
- Feasible-Space Sampling



Diagnose



Visual Integrity
SE(3) Alignment



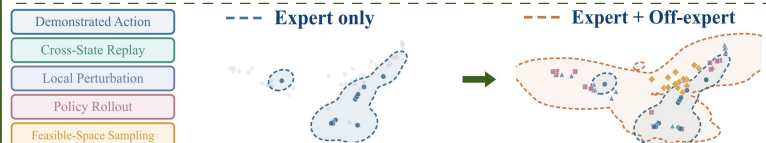
AC-WMs show an off-expert gap

2 IMPROVE



WorldSync

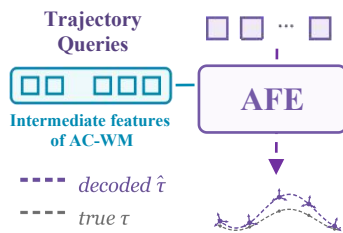
Action Coverage Expansion



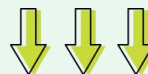
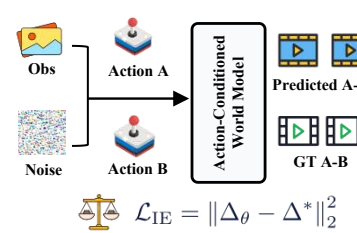
Multi-Task Simulation Data + Specific Task Real Data



Action-Forcing Expert



Intervention-Effect Supervision



WorldEcho Evaluation



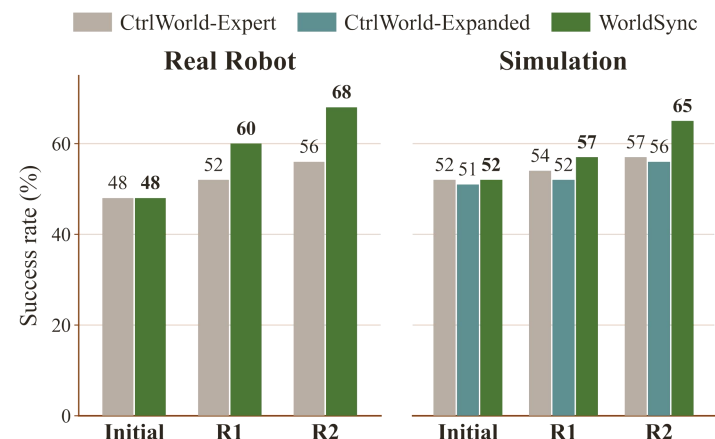
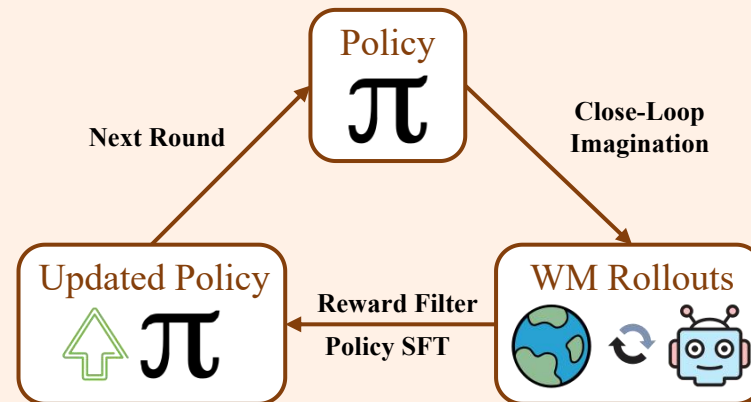
Visual Gate Pass $\uparrow$



Raw NDTW $\downarrow$

3 VALIDATE

Policy Improvement



Higher Policy Success Rate